

MP363 – Quantum Mechanics I

Problem Set 5

Due by the end of lecture on Friday, 18 December 2015

1. Consider a *classical* particle in a one-dimensional simple harmonic oscillator (SHO) potential. If the total energy of the particle is E , its kinetic energy is given by

$$\frac{1}{2}mv^2 = E - \frac{1}{2}m\omega^2 x^2.$$

Because the kinetic energy cannot be negative, the particle is thus constrained within the region $|x| \leq \sqrt{2E/m\omega^2}$.

The same is *not* true for a quantum-mechanical particle. To illustrate this, find the probability (expressed as a percentage) that a particle in the ground state of the QMSHO, whose energy is $E_0 = \hbar\omega/2$ and whose wavefunction is given by

$$\phi_0(t, x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \exp\left(-\frac{m\omega x^2}{2\hbar} - \frac{i\omega t}{2}\right),$$

is found in the classically forbidden region $|x| > \sqrt{2E_0/m\omega^2}$.

Note: if you do this correctly, your answer will involve a definite integral which *cannot* be computed analytically, but is in fact expressible in terms of what's called the “complementary error function”, denoted $\text{erfc}(x)$. **It is up to you** to do some research and find a value for this integral, either by numerically computing it yourself (via Simpson's rule or the like), using a program (like Matlab, Mathematica or Maple) or finding it in a mathematical functions reference (like the book by Abramowitz & Stegun or the Wolfram website). However you do it, your final answer should be expressed as a number in the form “XX.X%”.

2. Recall that if A is an operator on a Hilbert space $\mathcal{H}$ with inner product $\langle | \rangle$, then its adjoint $A^\dagger$ satisfies

$$\langle \psi | A | \zeta \rangle = \langle \zeta | A^\dagger | \psi \rangle^*$$

for all $\psi, \zeta \in \mathcal{H}$. We know that the creation and annihilation operators of the QMSHO act on an arbitrary wavefunction $\psi(x)$ as

$$\begin{aligned} a^\dagger \psi(x) &= \sqrt{\frac{m\omega}{2\hbar}} \left[x\psi(x) - \frac{\hbar}{m\omega} \frac{\partial \psi}{\partial x}(x) \right], \\ a \psi(x) &= \sqrt{\frac{m\omega}{2\hbar}} \left[x\psi(x) + \frac{\hbar}{m\omega} \frac{\partial \psi}{\partial x}(x) \right]. \end{aligned}$$

The inner product between any two wavefunctions ψ and ζ is

$$\langle \psi | \zeta \rangle = \int_{-\infty}^{\infty} \psi^*(x) \zeta(x) dx;$$

use this to show that $a^\dagger$ is indeed the adjoint of a . State *explicitly* any assumptions you make about the behaviour of the functions ψ and ζ in this proof, and justify them.

3. Recall that the creation and annihilation operators act on the eigenstates of the QMSHO as

$$a^\dagger |n\rangle = \sqrt{n+1} |n+1\rangle, \quad a |n\rangle = \sqrt{n} |n-1\rangle.$$

Use these, and the orthonormality of the eigenstates, to prove that the uncertainties in position and momentum for the state $|n\rangle$ are

$$\Delta x = \sqrt{\frac{\hbar}{m\omega} \left(n + \frac{1}{2} \right)} \quad \Delta p = \sqrt{m\hbar\omega \left(n + \frac{1}{2} \right)}.$$

(**Hint:** write X and P in terms of $a^\dagger$ and a and use those expressions to compute $\langle X \rangle$, $\langle X^2 \rangle$, $\langle P \rangle$ and $\langle P^2 \rangle$.)